

Santiago Patricio Gómez Ochoa



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PROFESSIONAL SUMMARY

Robotics and Digital Systems Engineering student with hands-on experience developing and integrating ROS 2-based robotic systems in Linux environments. Strong programming skills in C++ and Python, with practical experience in perception, navigation, and simulation using Gazebo and RViz2. Experienced in hardware-software integration across embedded platforms and NVIDIA Jetson devices, with a growing focus on hardware-accelerated robotics applications. Highly motivated to learn, collaborate, and contribute to high-performance robotics software development.

EDUCATION

Instituto Tecnológico y de Estudios Superiores de Monterrey

August 2023 - Currently

B.S. Robotics and Digital Systems Engineer CGPA - 9.65/10

SKILLS

Languages: Python, C++, C, JavaScript, Java, Assembly Language.

Technologies/Frameworks: OpenCV, FreeRTOS, TensorFlow, ROS 2 Jazzy/Humble, Micro-ROS, Raspbian, Nav2, PyTorch, Speech Brain, PX4, Gazebo, RViz2,.

Tools: VSCode, Arduino IDE, GitHub, MatLab, Raspberry Pi4.

Soft Skills: Team collaboration, leadership, collaborative development, logical and mathematical thinking, verbal communication skills, documentation skills, creativity, flexibility, project management, continuous learner.

PROJECTS

AutoModel Car | ROS2 Humble, Nvidia SDK, Jetson Orin Nano, Gazebo, RViz2, Micro-Ros **September 2025 - April 2026**

- Developed an autonomous scale vehicle using ROS 2, implementing perception, computer vision, control, and navigation nodes in C++/Python, validated in Gazebo and RViz2.
- Integrated ZED 2 stereo camera, RP LiDAR, and distributed microcontrollers software via micro-ROS, enabling real-time sensor fusion and low-latency control.
- Deployed perception pipelines on NVIDIA Jetson Orin Nano, exploring GPU-accelerated processing and system-level validation for obstacle avoidance and autonomous parking.
- Leveraged NVIDIA Jetson Orin Nano GPU acceleration for real-time perception pipelines, optimizing stereo vision and neural network inference under ROS 2.

Greenhouse Autonomous Assistant | ROS 2 Jazzy, PX4, RViz2, Gazebo, NAV2

September 2025

- Built an autonomous drone and differential robot using ROS 2 and PX4, implementing SLAM, Nav2-based planning, and LiDAR obstacle avoidance.
- Deployed an EfficientNet-B0 model for pest detection (>95% accuracy) with real-time image capture and wireless data transmission.

H.E.R.A. | ROS 2 Humble, RViz2, Gazebo, NAV2, Pytorch, SpeechBrain, SwiftUI, FastAPI

August 2025

- Designed an end-to-end robotic system where a mobile robot performs autonomous navigation routines and edge data collection using ROS 2 and Nav2.
- Integrated real-time NLP and voice AI pipelines with a FastAPI backend, demonstrating full robotic-software-service integration.

LANGUAGES

- Spanish—Native
- First Certificate in English—Level B2 Cambridge University
- DELF—Niveau A2 Alliance Française

COURSEWORK

- Advanced Embedded Systems - Tec de Monterrey 2025
- Intelligent Robotics Implementation - Tec de Monterrey 2026
- Robotics Foundation - Tec de Monterrey 2026